THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 9: **Symmetry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A wheel has 10 equal spokes. The smallest angle of rotation that fits the wheel back on itself is (degrees): 1

(a) 26 **(b)** 66 **(c)** 36 **(d)** 46

2. A regular polygon has each angle of rotation 12°. The number of lines of symmetry is: 1

(a) 31 **(b)** 29 **(c)** 32 **(d)** 30

3. The total angle through which a regular nonagon turns when rotated 4 times through its smallest angle (degrees): 1

(a) 190 **(b)** 170 **(c)** 160 **(d)** 150

4. A figure with order of rotational symmetry 1 but one line of symmetry is: 1

(a) equilateral triangle **(b)** square
(c) circle **(d)** isosceles triangle

5. A square is rotated about its centre through 70°. The smallest further turn (degrees) after which it again coincides with itself is: 1

(a) 10 **(b)** 110 **(c)** 30 **(d)** 20

6. The sum of the number of lines of symmetry of a regular hexagon and a regular pentagon is: 1

(a) 10 **(b)** 11 **(c)** 13 **(d)** 12

7. How many of the letters in P, Q, R, S, T have a line of symmetry? 1

(a) 1 **(b)** 2 **(c)** 0 **(d)** 3

8. A wheel has 12 equal spokes. The smallest angle of rotation that fits the wheel back on itself is (degrees): 1

(a) 60 **(b)** 30 **(c)** 20 **(d)** 40

9. How many of the letters in U, V, W, X, Y, Z have a line of symmetry? 1

(a) 4 **(b)** 7 **(c)** 5 **(d)** 6

10. A wheel has 15 equal spokes. The smallest angle of rotation that fits the wheel back on itself is (degrees): 1

(a) 14 **(b)** 34 **(c)** 54 **(d)** 24

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Draw (describe) the reflection of the triangle with vertices at x = 3, 5, 4 (heights 0, 0, 3) in a vertical mirror line at x = 0. Give the new x-coordinates. 2

12. A regular polygon rotates through 24° to coincide with itself. How many sides does it have? Name its order of symmetry. 2

13. Find the total number of lines of symmetry of a regular decagon and a regular hexagon together. 2

14. Draw (describe) the reflection of the triangle with vertices at x = 4, 6, 5 (heights 0, 0, 3) in a vertical mirror line at x = 0. Give the new x-coordinates. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A regular polygon has 10 lines of symmetry. Find its angle of rotation, the angle after 7 turns and the number of turns to make a full rotation. 3

16. Write the order of rotational symmetry and number of lines of symmetry for the letters X, Y, Z. Which have more lines than the order? 3

17. A clock face has 9 equally spaced marks. State the number of lines of symmetry (through marks) and the angle between neighbouring marks. If the minute hand moves from mark 0 to mark 2, through what angle has it turned? 3

OR

A regular polygon has 12 lines of symmetry. Find its angle of rotation, the angle after 5 turns and the number of turns to make a full rotation. 3

18. Write the order of rotational symmetry and number of lines of symmetry for the letters S, H, O. Which have more lines than the order? 3

OR

A clock face has 15 equally spaced marks. State the number of lines of symmetry (through marks) and the angle between neighbouring marks. If the minute hand moves from mark 0 to mark 4, through what angle has it turned? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A rangoli is a regular 10-pointed pattern. Find 5
(a) its lines of symmetry,
(b) the smallest rotation angle,
(c) how many rotations of that angle bring it back to the start,
(d) whether a rotation of 252° leaves it unchanged.

OR

Given figure types: a regular hexagon (lines 6) and a regular 12-gon (lines 12). Which has the larger angle of rotation? Find both angles, their difference and say which can fit into the other's symmetry (explain: 6 divides 12 or not). 5

20. A rangoli is a regular 12-pointed pattern. Find 5
(a) its lines of symmetry,
(b) the smallest rotation angle,
(c) how many rotations of that angle bring it back to the start,
(d) whether a rotation of 150° leaves it unchanged.

OR

Given figure types: a regular pentagon (lines 5) and a regular octagon (lines 8). Which has the larger angle of rotation? Find both angles, their difference and say which can fit into the other's symmetry (explain: 5 divides 8 or not). 5

END OF QUESTION PAPER — **Total: 40 Marks**